MTH203: Sequences and series
Lecture 10

In the previous lecture, we have proved that limits are well-behaved with respect to the operations of addition and multiplication. We will begin this lecture by proving the corresponding result for the operation of division (i.e. multiplication by the inverse). Then we will explore the notion of subsequences of a sequence and investigate their convergence.

1. Multiplicative inverses and limits

Theorem 1. *Let $(x_n)_{n\in\mathbb{N}}$ be a. convergent sequence in $\mathbb{R}$ such that $x_n \neq 0$ for all n and $\lim_{n\in\mathbb{N}} x_n \neq 0$. If $(y_n)_{n\in\mathbb{N}}$ is the sequence defined by $y_n := 1/x_n$ for all n, then $(y_n)_{n\in\mathbb{N}}$ is a convergent sequence and*

$$\lim_{n\in\mathbb{N}} y_n = \frac{1}{\lim_{n\in\mathbb{N}} x_n}.$$

Proof. Let $x = \lim_{n\in\mathbb{N}} x_n$ and let $\epsilon > 0$ be any given positive real number. We want to show that there exists an integer N such that for all $n > N$, we have $|(1/x) - (1/x_n)| < \epsilon$.

Informal discussion: The key idea is to use the following computation:

$$\begin{aligned} |(1/x) - (1/x_n)| &= \left|\frac{x_n - x}{x x_n}\right| \\ &= \left(\frac{1}{|x|}\right)\cdot\left(\frac{|x_n - x|}{|x_n|}\right) \end{aligned}$$

We would like to show that when n is large, the quantity $\left(\frac{1}{|x|}\right)\cdot\left(\frac{|x_n-x|}{|x_n|}\right)$ is small. The number $1/|x|$ is a fixed positive number, that does not depend on x. Thus, we essentially want to ensure that when n becomes very large, the quantity $\left(\frac{|x_n-x|}{|x_n|}\right)$ becomes small. We already know that the quantity $|x_n - x|$ becomes small as n becomes large. The situation is complicated by the fact that it is divided by $|x_n|$.

What if $|x_n|$ also becomes very small as n becomes large? That would perhaps prevent the number $\left(\frac{|x_n-x|}{|x_n|}\right)$ from becoming small. Now we have to recognize that $|x_n|$ cannot become arbitrarily small as n increases. Indeed, $\lim_{n\in\mathbb{N}} x_n = x$ is *non-zero*. So, the sequence $\{x_n\}_{n\in\mathbb{N}}$ eventually goes arbitrarily close to a number that is away from 0. Thus, after n is very large, we can ensure that $|x_n|$ is larger than some positive number, and so $1/|x_n|$ is less than some positive number. Thus, will allow us to conclude that $|x_n - x|\cdot(1/|x_n|)$ becomes small as n becomes large.

So, the key point is to use the fact that $\lim_{n\in\mathbb{N}} x_n$ is non-zero and use this to get some kind of bound on the quantity $1/|x_n|$ for all n. More precisely, we want to find some positive number M such that $1/|x_n| < M$ for all n. Then, if $|x_n - x|\cdot(1/|x_n|) < M\cdot|x_n - x|$. So, if we want to ensure that

$$\left(\frac{1}{|x|}\right)\cdot\left(\frac{|x_n - x|}{|x_n|}\right) < \epsilon,$$

it will suffice to ensure that

$$\left(\frac{1}{|x|}\right)\cdot M\cdot|x_n - x| < \epsilon,$$

which is equivalent to

$$|x_n - x| < \frac{|x| \cdot \epsilon}{M}.$$

This can always be arranged by making n sufficiently large.

We will now write this out formally.

Since $x \neq 0$, we have $|x| > 0$ and so $|x|/2 > 0$. Since $\lim_{n \to \mathbb{N}} x_n = x$, there exists N_1 such that if $n > N_1$, we have $|x - x_n| < |x|/2$. (Thus, when n is large enough, x_n stays "sufficiently close" to x and hence "sufficiently away" from 0.) Since $|x_n| + |x - x_n| \geqslant |x|$, we see that $|x_n| \geqslant |x| - |x - x_n|$. Thus, when $n > N_1$, we have $|x_n| \geqslant |x| - |x|/2 = |x|/2$. Thus, when $n > N_1$, we have $1/|x_n| < 2/|x|$.

Let $M = 1 + \max\left(\left\{1/|x_1|, 1/|x_2|, \ldots, 1/|x_{N_1}\right\} \cup \{2/|x|\}\right)$. Then, we can see that $1/|x_n| < M$ for all $n \in \mathbb{N}$. This is because, if $n < N_1$, we have

$$1/|x_n| \leqslant \max\left\{1/|x_1|, 1/|x_2|, \ldots, 1/|x_{N_1}|\right\} < M,$$

and if $n > N_1$, then we have

$$1/|x_n| < 2/|x| < M.$$

Thus, we have $1/|x_n| < M$ in either case.

Now, since $\lim_{n\in\mathbb{N}} x_n = x$, there exists an integer N such that if $n > N$, we have $|x_n - x| < \frac{|x|\cdot\epsilon}{M}$. Thus, if $n > N$, we have

$$M \cdot \left(\frac{|x_n - x|}{|x|}\right) < \epsilon.$$

Combining this with the inequality

$$\frac{|x_n - x|}{|x_n x|} = \left(\frac{1}{|x_n|}\right) \cdot \frac{|x_n - x|}{|x|} < M \cdot \left(\frac{|x_n - x|}{|x|}\right),$$

we obtain the inequality

$$\frac{|x_n - x|}{|x_n x|} < \epsilon$$

for $n > N$. This proves the result. □

This gives us the following result:

Corollary 2. *Let $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ be convergent sequences in $\mathbb{R}$. Assume that $y_n \neq 0$ for all $n \in \mathbb{N}$ and $\lim_{n\in\mathbb{N}} y_n \neq 0$.*

If $(z_n)_{n\in\mathbb{N}}$ is the sequence defined by $z_n := x_n/y_n$ for all n, then $(z_n)_{n\in\mathbb{N}}$ is a convergent sequence and

$$\lim_{n\in\mathbb{N}} z_n = \frac{\lim_{n\in\mathbb{N}} x_n}{\lim_{n\in\mathbb{N}} y_n}.$$

Proof. This corollary is an immediate consequence of Theorems 1 above and Theorem 5 from Lecture 9. □

2. Subsequences

Given a sequence $(x_n)_{n\in\mathbb{N}}$, a subsequence is essentially obtained by picking out certain terms from the sequence. The precise definition is as follows:

Definition 3. Let S be a set and let let $(x_n)_{n\in\mathbb{N}}$ be a sequence of elements of S. Then, if $(n_k)_{k\in\mathbb{N}}$ is a strictly increasing sequence of integers, the sequence $(x_{n_k})_{k\in\mathbb{N}}$ is said to be a subsequence of $(x_n)_{n\in\mathbb{N}}$.

If we view the sequence $(x_n)_{n\in\mathbb{N}}$ as a function $\mathbb{N} \to S$, $n \mapsto x_n$ and the sequence $(n_k)_{k\in\mathbb{N}}$ as a function function $\mathbb{N} \to \mathbb{N}$, $k \mapsto n_k$, then the subsequence $(x_n)_{n\in\mathbb{N}}$ is just the composition of these two functions. Informally, all we are doing is picking out some terms from the sequence $(x_n)_{n\in\mathbb{N}}$ (specifically, the n_1-term, then the n_2-term, and so on).

Example 4. Let $(x_n)_{n\in\mathbb{N}}$ be the sequence defined by $x_n = 1/n$. Then, the sequence $(y_n)_{n\in\mathbb{N}}$ where $y_n = 1/(2n+1)$ is a subsequence of $(x_n)_{n\in\mathbb{N}}$. This first few terms of this sequence are as follows:

$$\frac{1}{1}, \ \frac{1}{3}, \ \frac{1}{5}, \ \frac{1}{7}, \ \frac{1}{9}, \ \frac{1}{11}, \ \ldots$$

However, now consider the sequence $(z_n)_{n\in\mathbb{N}}$ defined by $z_n = \frac{1}{2n-1-2\cdot(-1)^n}$. You can verify that the first few terms are as follows:

$$\frac{1}{3}, \ \frac{1}{1}, \ \frac{1}{7}, \ \frac{1}{5}, \ \frac{1}{11}, \ \frac{1}{9}, \ \ldots$$

This is *not* a subsequence of the sequence $(x_n)_{n\in\mathbb{N}}$ despite the fact that all the terms of $(z_n)_{n\in\mathbb{N}}$ occur in $(x_n)_{n\in\mathbb{N}}$. This is because the order of the terms is being changed.

As the example above shows, for $(x_{n_k})_{k\in\mathbb{N}}$ to be a subsequence of $(x_n)_{n\in\mathbb{N}}$, it is crucial that $n_1, n_2, \ldots$ should be a *strictly increasing* sequence of positive integers. Here is a small and rather obvious lemma. (You may use this lemma directly without quoting it from any specific source since it is very obvious. I am writing it out here just for the sake of clarity.)

Lemma 5. *Let $(n_k)_{k\in\mathbb{N}}$ be a strictly increasing sequence in $\mathbb{N}$. Then, for any k, we have $n_k \geqslant k$.*

Proof. We prove the result by induction on k. The result is clearly obvious for $k = 1$. Suppose it is known that $n_k \geqslant k$ for some integer $k \geqslant 1$.

We will show that $n_{k+1} \geqslant k+1$. Indeed, since the sequence is strictly increasing, we have $n_{k+1} > n_k$. Thus, since this is a sequence of integers, $n_{k+1} \geqslant n_k + 1$. By the induction hypothesis, $n_k \geqslant k$ and so $n_k + 1 \geqslant k + 1$. Thus, putting the two inequalities together, we get $n_{k+1} \geqslant k + 1$. This completes the induction step. □

Here is a very basic theorem about convergence of subsequences:

Theorem 6. *Let $(x_n)_{n\in\mathbb{N}}$ be a convergent sequence in $\mathbb{R}$. Let $(x_{n_k})_{k\in\mathbb{N}}$ be a subsequence of $(x_n)_{n\in\mathbb{N}}$. Then, $(x_{n_k})_{k\in\mathbb{N}}$ is a convergent sequence and*

$$\lim_{k\to\infty} x_{n_k} = \lim_{n\to\infty} x_n.$$

Proof. Let $\epsilon > 0$ be given. Let $\lim_{n\to\infty} x_n$ be denoted by x. Since $(x_n)_{n\in\mathbb{N}}$ converges to x, there exists an integer N such that for any $n > N$, we have $|x - x_n| < \epsilon$.

Now, suppose $k > N$. Then, since $n_k \geqslant k$, we also have $n_k > N$. Thus, $|x - x_{n_k}| < \epsilon$. This shows that $(x_{n_k})_{k\in\mathbb{N}}$ converges to x. □

Corollary 7. *Suppose a sequence $(x_n)_{n\in\mathbb{N}}$ has a divergent subsequence. Then $(x_n)_{n\in\mathbb{N}}$ is also a divergent sequence.*

Proof. This is an immediate consequence of the above theorem. □

3. Bolzano-Weierstrass theorem

We have seen in Lecture 8 that any convergent sequence in a metric space is bounded. However, we also saw that the converse is not true – it is possible for a bounded sequence to be divergent. However, we do have the following partial converse for real numbers.

Theorem 8 (Bolzano-Weierstrass theorem)**.** *Any bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

Proof. Let $(x_n)_{n\in\mathbb{N}}$ be a bounded sequence in $\mathbb{R}$. Thus, there exists a positive real number M such that $|x_n| < M$ for all n. Thus, $x_n \in [-M, M]$ for all n.

We will now construct a sequence of closed intervals $I_1, I_2, \ldots$ such that they have the following properties:

(a) $I_j \subset I_{j-1}$ for all positive integers $j > 1$.
(b) There exist infinitely many integers n such that $x_n \in I_j$. (We may *informally* express this by saying that I_j contains infinitely many terms of the sequence $(x_n)_{n\in\mathbb{N}}$.
(c) If $I_k = [a_k, b_k]$, then $b_k - a_k = M/2^{k-2}$.

> *Comment:* Observe as k becomes larger and larger, the number $M/2^{k-2}$ will get very small. The number $b_k - a_k$ is the "length" of the interval I_k. Thus, condition (c) is saying that the length of I_k becomes very small as k increases.

We will construct this sequence by an inductive process. We define I_1 to be the interval $[-M, M]$. Since this interval already contains all the x_n's, it clearly satisfies condition (b). Also, we have $a_1 = -M$ and $a_2 = M$. So $b_1 - a_1 = 2M = M/2^{-1} = M/2^{1-2}$. Thus, condition (c) also holds. (We do not have to check condition (a) for this interval since that only applies for integers $j > 1$.)

Now, suppose that k is an integer $\geqslant 1$ and $I_1, I_2, \ldots, I_k$ have been chosen so that they satisfy conditions (a), (b) and (c). We wish to construct the interval I_{k+1}.

Let $I_k = [a_k, b_k]$. We divide this interval into two equal halves. Let I'_{k+1} be the interval $[a_k, \frac{a_k+b_k}{2}]$ and $I''_{k+1} = [\frac{a_k+b_k}{2}, b_k]$. Thus, we have $I'_{k+1} \cup I''_{k+1} = I_k$ and $I'_{k+1} \cap I''_{k+1} = \{\frac{a_k+b_k}{2}\}$.

Since the interval I_k contains infinitely many terms of the sequence $(x_n)_{n\in\mathbb{N}}$, at least one of the intervals I'_{k+1} and I''_{k+1} contains infinitely many terms of this sequence. We will call this interval I_{k+1}. This interval clearly satisfies conditions (a) and (b) above. If $I_{k+1} = I'_{k+1}$, then $b_{k+1} = \frac{a_k+b_k}{2}$ and $a_{k+1} = a_k$. Thus,

$$b_{k+1} - a_k = \frac{b_k - a_k}{2} = \frac{M \times 2^{k-2}}{2} = \frac{M}{2^{(k+1)-2}}.$$

A similar calculation shows that even in the case $I_{k+1} = I''_{k+1}$, we have $b_{k+1} - a_{k+1} = M/2^{(k+1)-2}$. Thus, the interval I_{k+1} also satisfies condition (c). This completes the induction. Using this inductive process, we are able to construct the required sequence of closed intervals.

We have seen in Lecture 5 that a nested sequence of closed intervals has a non-empty intersection. Thus, $\bigcap_{k=1}^{\infty} I_k \neq \varnothing$. We will show that the intersection has exactly one element.

Suppose x and y are *distinct* elements of $\bigcap_{k=1}^{\infty} I_k$. Then, $|x - y| > 0$. Given any real number x, there exists a positive integer N such that if $m > N$, we have $2^m > x$. (The proof for this is essentially the same as the one in Lecture 6, Lemma 1 where we proved a similar statement for powers of 10). We will prove a general statement like this for powers of any real number greater than 1 in Lecture 12.) Thus, there exists an integer N such that if $m > N$, we have $2^m > \frac{4M}{|x-y|}$ and so $|x - y| > \frac{4M}{2^m} = \frac{M}{2^{m-2}}$. Thus, if $m > M$, we have

$$b_m - a_m = \frac{M}{2^{m-2}} < |x - y|.$$

However, $x, y \in I_m$. So $|x - y| < b_m - a_m$, which gives us a contradiction. This shows that $\bigcap_{k=1}^{\infty} I_k$ cannot contain two *distinct* elements. Thus, it is a singleton set and we denote its only element by x.

We now show that there exists a subsequence of $(x_n)_{n\in\mathbb{N}}$ which converges to x.

We first choose an increasing sequence of integers $n_1 < n_2 < \ldots$ such that $x_{n_k} \in I_k$. This sequence is constructed inductively. Choose $n_1 = 1$. We already know that $x_1 \in [-M, M] = I_1$. Now, suppose $n_1 < n_2 < \cdots < n_k$ have been chosen such that $x_{n_j} \in I_j$ for $1 \leqslant j \leqslant k$. Recall that there are infinitely many integers n such that $x_n \in I_{k+1}$. Thus, we can choose an integer $n_{k+1} > n_k$ such that $x_{n_{k+1}} \in I_{k+1}$. This completes the inductive step of the construction.

We claim that $\lim_{k\to\infty} x_{n_k} = x$. Indeed, let $\epsilon > 0$ be given. Now, choose N such that for any $m > N$, we have $2^m > \frac{4M}{\epsilon}$. Thus, if $m > N$, we have $\epsilon > \frac{M}{2^{m-2}} = b_m - a_m$. Since both x and x_{n_m} are elements of I_m, $|x - x_{n_m}| < b_m - a_m < \epsilon$.

This shows that $\lim_{k\to\infty} x_{n_k} = x$. ☐

4. Reading suggestions

The result in Section 1 of this lecture appears as part of Theorem 2.3.3 in the textbook ([1]) and the rest of the material appears in Section 2.5.

References

[1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. 5